New features of rtauargus

a step further in the automation of the analysis

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Outline

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01

Recap on the analysis of a publication

RECAP ON THE ANALYSIS OF A PUBLICATION

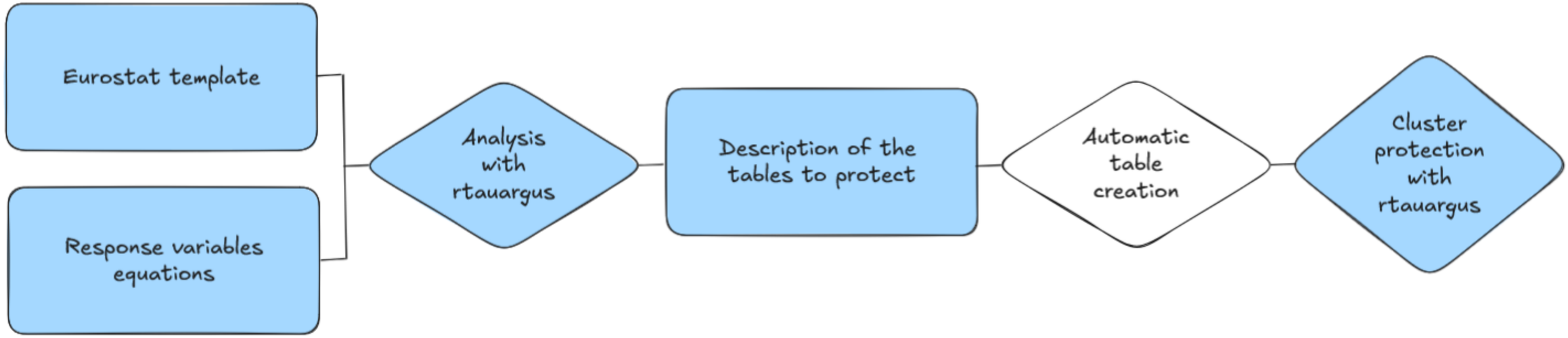


Table protection workflow

1.1 Importing Eurostat template for EBSBDS[1] ✍️

- ✍️ Manual steps with R
- ⚙️ Package functions

▶ Code

▶ Code

	FREQ <chr>	REF_AREA <chr>	TABLENAME <chr>	INDICATOR <chr>	ACTIVITY <chr>	NUMBER_EMPL <chr>	LEGAL_FORM <chr>	SURVIVAL_AGE <chr>
1	A	FR	T11	EMPL	B	E0	_T	_Z
2	A	FR	T11	EMPL	B	E1T4	_T	_Z
3	A	FR	T11	EMPL	B	E5T9	_T	_Z
4	A	FR	T11	EMPL	B	EGE10	_T	_Z
5	A	FR	T11	EMPL	B	_T	LL	_Z
6	A	FR	T11	EMPL	B	_T	PA	_Z

6 rows | 1-9 of 21 columns

1.2 Selecting the columns needed for the analysis ✍️

- Code

```
TIME_PERIOD INDICATOR ACTIVITY NUMBER_EMPL LEGAL_FORM
       2022       SAL        B          E0         _T
       2022       SAL        B        E1T4         _T
       2022       SAL        B        E5T9         _T
       2022       SAL        B       EGE10         _T
       2022       SAL        B          _T         LL
       2022       SAL        B          _T         PA
```

SAL: number of employees
SAL_DTH: number of employees in enterprise deaths

RECAP ON THE ANALYSIS OF A PUBLICATION

1.3 Going from the template to the description of the published tables ⚙️

▸ Code

```
treating the field 2021
treating the field 2022
```

▸ Code

table_name <chr>	field <chr>	indicator <chr>	spanning_1 <chr>	spanning_2 <chr>	hrc_spanning_1 <chr>
table_2021_SAL_DTH_1	2021	SAL_DTH	ACTIVITY	LEGAL_FORM	hrc_activity_131
table_2021_SAL_DTH_2	2021	SAL_DTH	ACTIVITY	NUMBER_EMPL	hrc_activity_131
table_2022_SAL_1	2022	SAL	ACTIVITY	LEGAL_FORM	hrc_activity_131
table_2022_SAL_2	2022	SAL	ACTIVITY	NUMBER_EMPL	hrc_activity_131
table_2022_SAL_DTH_1	2022	SAL_DTH	ACTIVITY	LEGAL_FORM	hrc_activity_131
table_2022_SAL_DTH_2	2022	SAL_DTH	ACTIVITY	NUMBER_EMPL	hrc_activity_131

6 rows | 1-6 of 7 columns

The 22 572 rows of the template actually represented 6 tables to publish.

1.4 Adding the hierarchies for field and indicator ✍️

▸ Code

	table_name <chr>	field <chr>	indicator <chr>	spanning_1 <chr>	spanning_2 <chr>	hrc_spanning_1 <chr>
1	table_2021_SAL_DTH_1	2021	SAL_DTH	ACTIVITY	LEGAL_FORM	hrc_activity_131
2	table_2021_SAL_DTH_2	2021	SAL_DTH	ACTIVITY	NUMBER_EMPL	hrc_activity_131
3	table_2022_SAL_1	2022	SAL	ACTIVITY	LEGAL_FORM	hrc_activity_131
4	table_2022_SAL_2	2022	SAL	ACTIVITY	NUMBER_EMPL	hrc_activity_131
5	table_2022_SAL_DTH_1	2022	SAL_DTH	ACTIVITY	LEGAL_FORM	hrc_activity_131
6	table_2022_SAL_DTH_2	2022	SAL_DTH	ACTIVITY	NUMBER_EMPL	hrc_activity_131

6 rows | 1-7 of 10 columns

1.5 Analysing the links between tables ⚙️

▸ Code

cluster <chr>	table_name <chr>	field <chr>	indicator <chr>	spanning_1 <chr>	spanning_2 <chr>
2021.SAL_DTH	table_2021_SAL_DTH_1	2021	SAL_DTH	HRC_ACTIVITY_131	HRC_LEGAL_FORM_3
2021.SAL_DTH	table_2021_SAL_DTH_2	2021	SAL_DTH	HRC_ACTIVITY_131	HRC_NUMBER_EMPL_4
2022.SAL	table_2022_SAL_1	2022	SAL	HRC_ACTIVITY_131	HRC_LEGAL_FORM_3
2022.SAL	table_2022_SAL_2	2022	SAL	HRC_ACTIVITY_131	HRC_NUMBER_EMPL_4
2022.SAL_DTH	table_2022_SAL_DTH_1	2022	SAL_DTH	HRC_ACTIVITY_131	HRC_LEGAL_FORM_3
2022.SAL_DTH	table_2022_SAL_DTH_2	2022	SAL_DTH	HRC_ACTIVITY_131	HRC_NUMBER_EMPL_4

6 rows | 1-6 of 8 columns

The analysis shows that 6 tables need to be protected in 3 different clusters.

02

New features of rtauargus

2.1 New features

New parameter added to `analyse_metadata()` allowing the user to specify links between response variables (version **1.3.2**).

Fix bug due to encoding, french months with accents such as “Décembre” or “Février” (version **1.3.3**).

2.2 Example of links between response variables

Table 1. Active enterprises

	0–9	10+	Total
North	120	30	150
South	100	23	123
Total	220	53	273

Table 2. Enterprise births

	0–9	10+	Total
North	20	5	25
South	18	2	20
Total	38	7	45

Table 3. Surviving enterprises

	0–9	10+	Total
North	100	25	125
South	82	21	103
Total	182	46	228

Note: Active enterprises = Enterprise births + Surviving enterprises.

NEW FEATURES OF RTAUARGUS

2.3 Protecting tables with links between response variables

GEO	SIZE CLASS	ACTIVE	FREQ
North	0-9	birth	20
North	10+	birth	5
North	0-9	surviving	100
North	10+	surviving	25
North	0-9	active	120
North	10+	active	30
South	0-9	birth	18
South	10+	birth	2
…	…	…	
Total	Total	active	275

03

Use case: Annual survey on industrial energy consumption

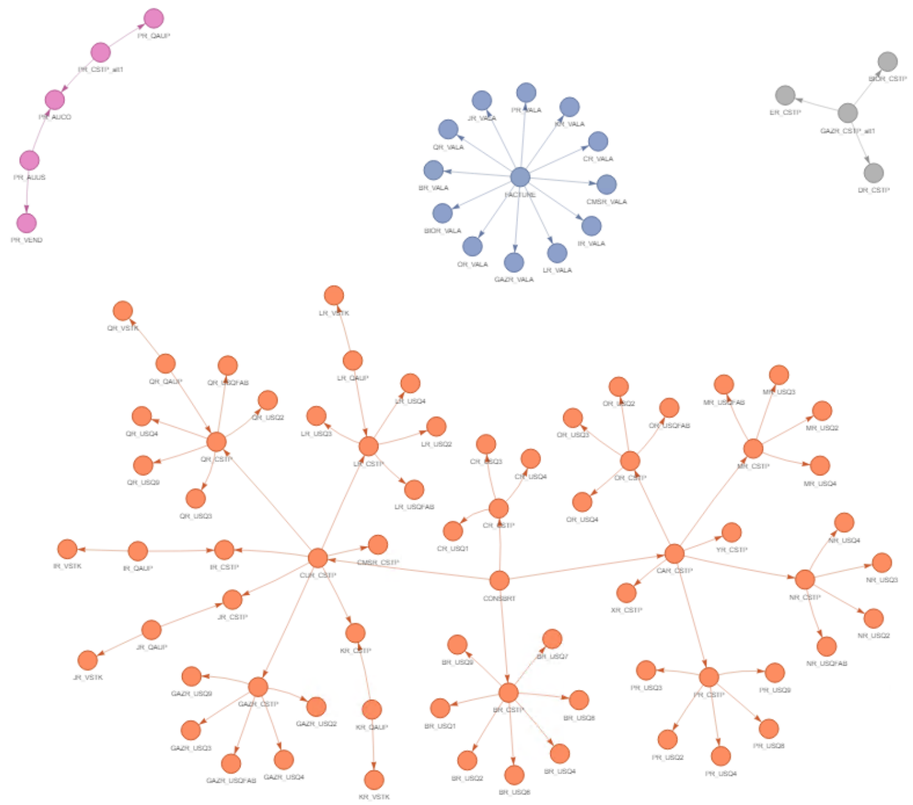
3.1 From crossings to published tables

Extract of the French template, one row equals one cell (same as Eurostat template).

3.2 Equations of response variable

eq_name	eq_indicator	unit
eq_1	CONSBRT = BR_CSTP + CAR_CSTP + …	TOE
eq_2	BR_CSTP = BR_USQ1 + BR_USQ2 + …	TOE
eq_6	FACTURE = BIOR_VALA + BR_VALA + …	EUR
eq_17	BR_PROD = BR_THER + BR_NTHE	GWH

3.3 Links between indicators



Open interactive graph

3.4 Result

cluster	table	field	indicator	span_1	span_2	hrc_1	hrc_2
alpha	T1	2023	EUR	GEO	eq6	hrc_geo13	hrc_eq6
beta	T2	2023	TOE	GEO	eq1_eq2_...	hrc_geo13	hrc_eq1_eq2_...
beta	T3	2023	TOE	GEO	eq75	hrc_geo13	hrc_eq75
...	...	...	...	...	...	...	...

– 72 tables to publish by region (218 in the full publication)

– 9 tables to protect

– 6 clusters

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